Troubleshooting guide

Symptom → likely cause → fix

The engineer's method: 1. Describe exactly what happens. 2. Guess one cause. 3. Change one thing. 4. Test and write it down.

What you see	Likely cause	Try this
Nothing happens at all	Battery flat or switch off; upload failed	Charge/switch on; re-upload and read any red errors
Motors twitch or act randomly	No common ground	Connect L298N GND to Arduino GND
Robot spins in circles	One motor wired backwards	Swap that motor's two wires at the L298N; test with motor_test.ino
Robot drives backwards	Both motors reversed	Swap both motors' wires, or swap IN1/IN2 and IN3/IN4
Turns away from the line	Left and right swapped (sensors or rules)	Swap the OUT wires on D2/D3, or swap the two turn rules
Stops on plain white	Sensors too sensitive or too high	Turn the knobs; lower sensors to 5–10 mm; check ON_LINE
Misses sharp corners	Turning too gently or too fast	Lower BASE_SPEED; set TURN_SPEED to 0 or −80 (pivot)
Wobbles a lot on straights	Too fast or sensors too far apart	Lower BASE_SPEED; move sensors to about 3 cm apart
Upload fails	Wrong board or port; robot powered from battery	Tools → Board: Arduino Uno; Tools → Port; try another USB port
Works, then stops after a while	Batteries running low	Charge the batteries

Ask the robot what it sees

Upload `ir_sensor_test.ino`, open the Serial Monitor at 9600 baud and slide the robot across the line. Each sensor should say BLACK only over the tape.